PORTFOLIO OPTIMIZATION & INVESTMENT STRATEGY REPORT

10-Year vs. 30-Year Wealth Generation Models

Siddharth Kusuma
M.S. in Engineering Management
The George Washington University
(G26921757)

"The essence of investment management is the management of risks, not the management of returns."
— Benjamin Graham (Father Of Value Investing)

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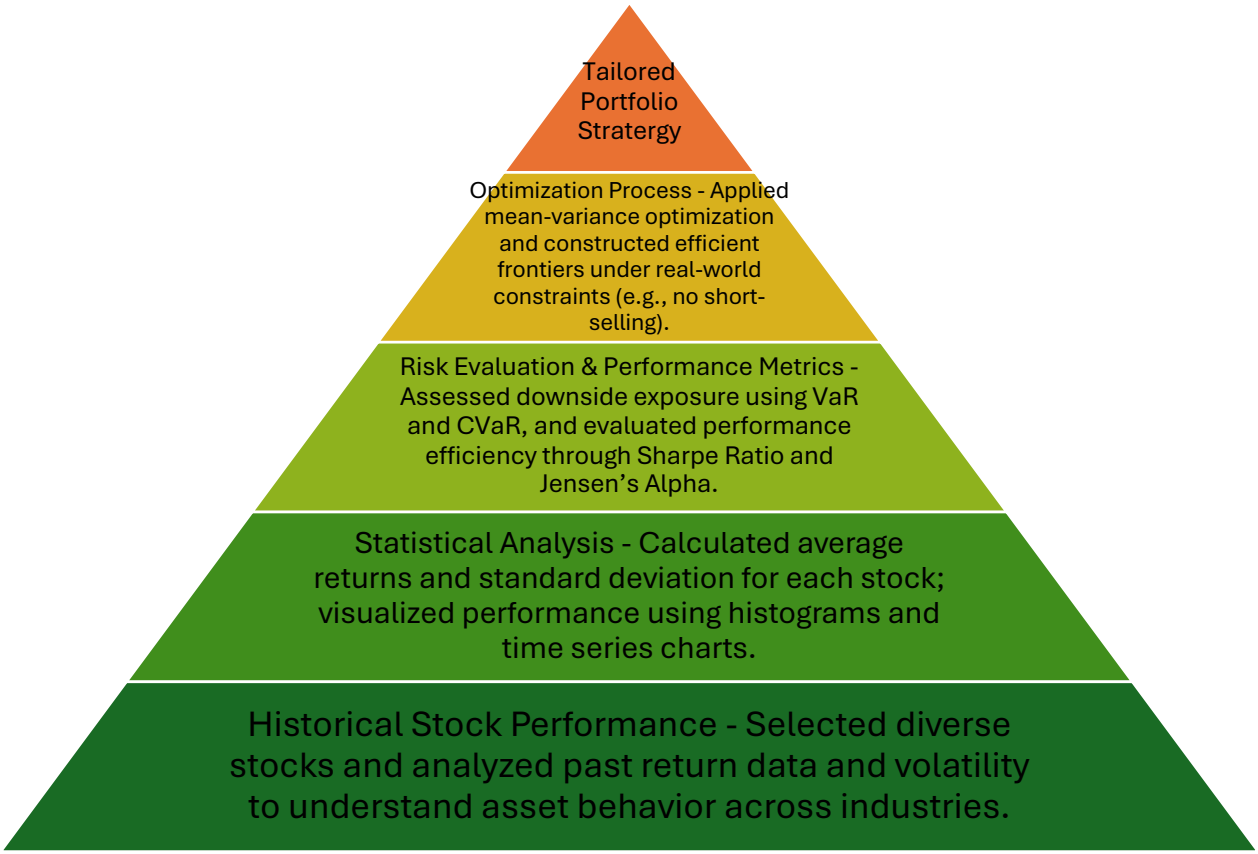
Introduction

Building a strong investment portfolio requires more than just selecting high-performing assets—it involves balancing return potential with risk, considering investor goals, and applying data-driven techniques to guide allocation decisions. This report aims to construct and analyze two portfolios: one designed for a 10-year time horizon, and another for a 30-year horizon.

The objective is to optimize returns for two different investor profiles, each with its own risk tolerance and financial goals. To achieve this, we apply a range of quantitative tools including mean-variance optimization, the Sharpe Ratio, Jensen's Alpha, and risk assessment metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). These tools help evaluate expected return, volatility, and potential downside exposure for various asset combinations.

The core of this process begins with selecting a group of stocks from diverse industries including technology, finance, industrials, and e-commerce. Each stock is evaluated for its return behavior and contribution to diversification. The insights derived from this analysis inform the portfolio construction process and allow us to build strategies suited to medium- and long-term investment objectives.

To design meaningful portfolio strategies, it's important to follow a structured process that builds from foundational data toward actionable insights. The approach used in this report relies on progressively layering statistical analysis, risk assessment, and optimization techniques to arrive at a well-balanced investment strategy. The following visual summarizes this multi-step framework, illustrating how each layer of analysis supports the development of the final portfolio recommendations.



Before diving into the analytical framework and portfolio design, the following section offers a brief overview of the selected stocks and their relevance in shaping the strategies that follow.

Stocks

1. NVIDIA (NVDA)

About the Company:

NVIDIA is a technology company best known for making high-performance graphics cards (GPUs), which are used in gaming, professional design, data centers, and increasingly in artificial intelligence (AI). The company has become a central player in the development of AI technologies, with its chips powering everything from research labs to self-driving cars.

Trading Performance:

NVIDIA's stock has seen a significant rise in recent years, especially in 2023 and 2024, due to increased global demand for AI-related hardware. It became one of the top-performing stocks in the S&P 500 and reached all-time highs in early 2024. Investors have shown strong confidence in its future, leading to high trading volumes and valuation growth.

Expected Future:

NVIDIA is expected to continue growing as AI adoption increases across industries. With strong demand for data center GPUs and continued innovation in chip design, many analysts believe the company will remain at the forefront of the tech world. While the stock may be volatile due to high expectations, its long-term outlook remains positive.

(**NVDIA** Chart with each candle representing 1 month)

2. Apple Inc. (AAPL)

About the Company:

Apple Inc. is a leading technology company known for its innovative consumer electronics, software, and services. Its product lineup includes the iPhone, iPad, Mac computers, Apple Watch, and services like the App Store, Apple Music, and iCloud. Apple is headquartered in Cupertino, California, and has a strong global presence.

Trading Performance:

In 2024, Apple's stock demonstrated robust performance, achieving a total return of 30.71%. The stock reached an all-time high closing price of $258.74 on December 26, 2024.

Expected Future:

Looking ahead, Apple plans to invest over $500 billion in the U.S. over the next four years, focusing on research and development in areas like custom silicon, hardware engineering, software development, and artificial intelligence. Analysts project continued growth, with stock price forecasts for 2025 ranging between $180 and $249, reflecting confidence in Apple's sustained innovation and market leadership.

(**APPLE** Chart with each candle representing 1 month)

3. Amazon.com Inc. (AMZN)

About the Company:

Amazon is a global technology and e-commerce leader headquartered in Seattle, Washington. It began as an online bookstore but has since expanded into various sectors, including cloud computing (Amazon Web Services), digital streaming, artificial intelligence, and logistics. Amazon's diversified business model allows it to serve both consumers and businesses worldwide.

Trading Performance:

In 2024, Amazon's stock experienced significant growth, closing the year at $219.39, marking a 44.39% increase from the previous year. The stock reached an all-time high closing price of $242.06 on February 4, 2025. However, as of April 23, 2025, the stock has adjusted to $180.53, reflecting a 17.71% decrease year-to-date.

Expected Future:

Analysts anticipate that Amazon's revenue will grow by approximately 10% in 2025, driven by continued expansion in its e-commerce and cloud computing segments. However, potential challenges such as tariff concerns and increased capital expenditures may impact profitability. Despite these factors, the consensus among analysts remains positive, with an average 12-month price target of $250.20, suggesting a potential upside from current levels.

(**AMAZON** Chart with each candle representing 1 month)

4. Super Micro Computer, Inc. (SMCI)

About the Company:

Super Micro Computer, Inc., commonly known as Supermicro, is a U.S.-based technology company specializing in high-performance server and storage solutions. Founded in 1993 and headquartered in San Jose, California, Supermicro designs and manufactures a wide range of computing products, including server systems, motherboards, and networking devices. The company is recognized for its focus on energy-efficient computing and its ability to rapidly deliver customized solutions to data centers, cloud computing, enterprise IT, and big data markets.

Trading Performance:

In 2024, Supermicro's stock experienced significant volatility. The stock reached an all-time high of $122.90 on March 8, 2024, driven by strong demand for AI-optimized servers and robust financial performance. However, the stock faced pressure later in the year due to concerns over accounting practices and regulatory scrutiny, leading to a decline in share price. As of April 23, 2025, SMCI is trading at approximately $32.90, reflecting a substantial decrease from its peak.

Expected Future:

Looking ahead, Supermicro aims to capitalize on the growing demand for AI and cloud computing infrastructure. The company has projected fiscal year 2025 revenue between $23.5 billion and $25 billion, with aspirations to reach $40 billion in revenue by fiscal year 2026. Analysts have a consensus 12-month price target of $52.57 for SMCI, indicating potential upside from current levels. However, investors should remain cautious due to ongoing regulatory investigations and the company's efforts to strengthen internal controls and corporate governance.

(**Super Micro Computer, Inc** Chart with each candle representing 1 month)

5. American Express Company (AXP)

About the Company:

American Express is a global financial services corporation headquartered in New York City. Founded in 1850, the company is renowned for its charge and credit card products, travel services, and financial offerings for both consumers and businesses. Operating in over 130 countries, American Express has built a reputation for catering to premium customers and delivering exceptional customer service.

Trading Performance:

In 2024, American Express's stock demonstrated robust performance, closing the year at $295.11, marking a 61.1% increase from the previous year. The stock reached an all-time high closing price of $324.79 on January 23, 2025. However, as of April 23, 2025, the stock has adjusted to $260.14, reflecting a 9.42% decrease year-to-date.

Expected Future:

Looking ahead, analysts anticipate that American Express will continue its growth trajectory. The company has projected revenue growth between 8% and 10% for 2025, with earnings per share expected to increase by 12% to 16%. Analyst price targets for AXP in 2025 range from $267.32 to $341.49, indicating potential upside from current levels. The company's focus on premium customers and its strong brand position it well for sustained performance.

(**American Express Company** Chart with each candle representing 1 month)

6. Goldman Sachs Group Inc. (GS)

About the Company:

Goldman Sachs is a leading global investment banking and financial services firm headquartered in New York City. Founded in 1869, the company offers a wide range of services, including investment banking, securities trading, asset management, and consumer banking. Goldman Sachs is renowned for advising major corporations, governments, and high-net-worth individuals, playing a pivotal role in global financial markets.

Trading Performance:

In 2024, Goldman Sachs's stock experienced significant growth, closing the year at $569.80, marking a 52.03% increase from the previous year. The stock reached an all-time high closing price of $668.87 on February 18, 2025. However, as of April 24, 2025, the stock has adjusted to $545.37, reflecting a 4.29% decrease year-to-date.

Expected Future:

Looking ahead, Goldman Sachs anticipates an increase in capital raising and mergers and acquisitions (M&A) activities in 2025, attributing this expected growth to the new U.S. administration's policies. However, the firm also acknowledges potential challenges, including policy uncertainties and trade tensions, which could impact market confidence and financial conditions.

(**Goldman Sachs** Chart with each candle representing 1 month)

7. Honeywell International Inc. (HON)

About the Company:

Honeywell International Inc. is a diversified industrial conglomerate headquartered in Charlotte, North Carolina. The company operates across various sectors, including aerospace, building technologies, performance materials, and safety solutions. Honeywell is known for integrating advanced software and technology to drive innovation, operational efficiency, and energy optimization across industries.

Trading Performance:

In 2024, Honeywell's stock reached an all-time high closing price of $234.74 on December 16. However, as of April 24, 2025, the stock has adjusted to $200.74, reflecting a 10.66% decrease year-to-date. Over the past 52 weeks, the stock has experienced a range between $179.36 and $242.77.

Expected Future:

Looking ahead, Honeywell plans to restructure into three independent companies focusing on automation, aerospace, and advanced materials, aiming to enhance shareholder value. For fiscal year 2025, the company projects sales between $39.6 billion and $40.6 billion, with adjusted earnings per share ranging from $10.10 to $10.50. Analysts forecast annual revenue growth of approximately 4.7% and earnings growth of 7.9%, indicating a cautiously optimistic outlook.

(**Honeywell International Inc** Chart with each candle representing 1 month)

8. Microsoft Corporation (MSFT)

About the Company:

Microsoft is a global technology leader headquartered in Redmond, Washington. Founded in 1975, the company develops, licenses, and supports a wide range of software products, services, and devices. Its flagship products include the Windows operating system, Microsoft Office suite, and Azure cloud services. Microsoft also owns LinkedIn, GitHub, and Xbox, and has been investing heavily in artificial intelligence (AI) and cloud computing to drive future growth.

Trading Performance:

In 2024, Microsoft's stock reached an all-time high closing price of $464.85 on July 5. However, as of April 24, 2025, the stock has adjusted to $387.30, reflecting a 7.93% decrease year-to-date. The 52-week range for MSFT is between $344.79 and $468.35, with an average price of $418.33 over the past year.

Expected Future:

Looking ahead, analysts anticipate that Microsoft's stock will trade between $364.72 and $519.65 in 2025, with an average annualized price of $445.18. This projection suggests a potential return on investment of approximately 34.17% compared to current levels. However, some analysts have expressed caution due to macroeconomic uncertainties and slower revenue growth in certain segments. For instance, Citi Research analysts recently lowered their price target from $497 to $480, citing moderating expectations amid a global economic slowdown. Despite these concerns, the consensus among analysts remains optimistic, with 55 out of 61 analysts rating Microsoft as a "Buy" ahead of its upcoming fiscal third-quarter earnings report.

(**Microsoft Corporation** Chart with each candle representing 1 month)

9. Bank of America Corporation (BAC)

About the Company:

Bank of America is one of the largest financial institutions in the United States, offering a wide range of banking, investment, asset management, and other financial and risk management products and services. Headquartered in Charlotte, North Carolina, the bank serves individual consumers, small and middle-market businesses, and large corporations with a full suite of banking services, including checking and savings accounts, loans, credit cards, and wealth management.

Trading Performance:

As of April 24, 2025, Bank of America's stock closed at $39.58. The stock reached an all-time high closing price of $47.44 on February 6, 2025. Over the past 52 weeks, the stock has fluctuated between a low of $33.06 and a high of $48.08, with an average price of $41.79.

Expected Future:

Analysts have a consensus rating of "Moderate Buy" for Bank of America, with an average 12-month price target of $47.20, indicating potential for price appreciation. UBS analyst Erika Najarian has identified Bank of America as an undervalued opportunity, upgrading the stock to "Buy" with a 12-month price target of $53, citing potential benefits from deregulation and higher interest rates. Additionally, forecasts suggest that Bank of America's stock could reach up to $50.31 by October 2025, representing a potential return on investment of 27.11% from current levels.

(**Bank Of America** Chart with each candle representing 1 month)

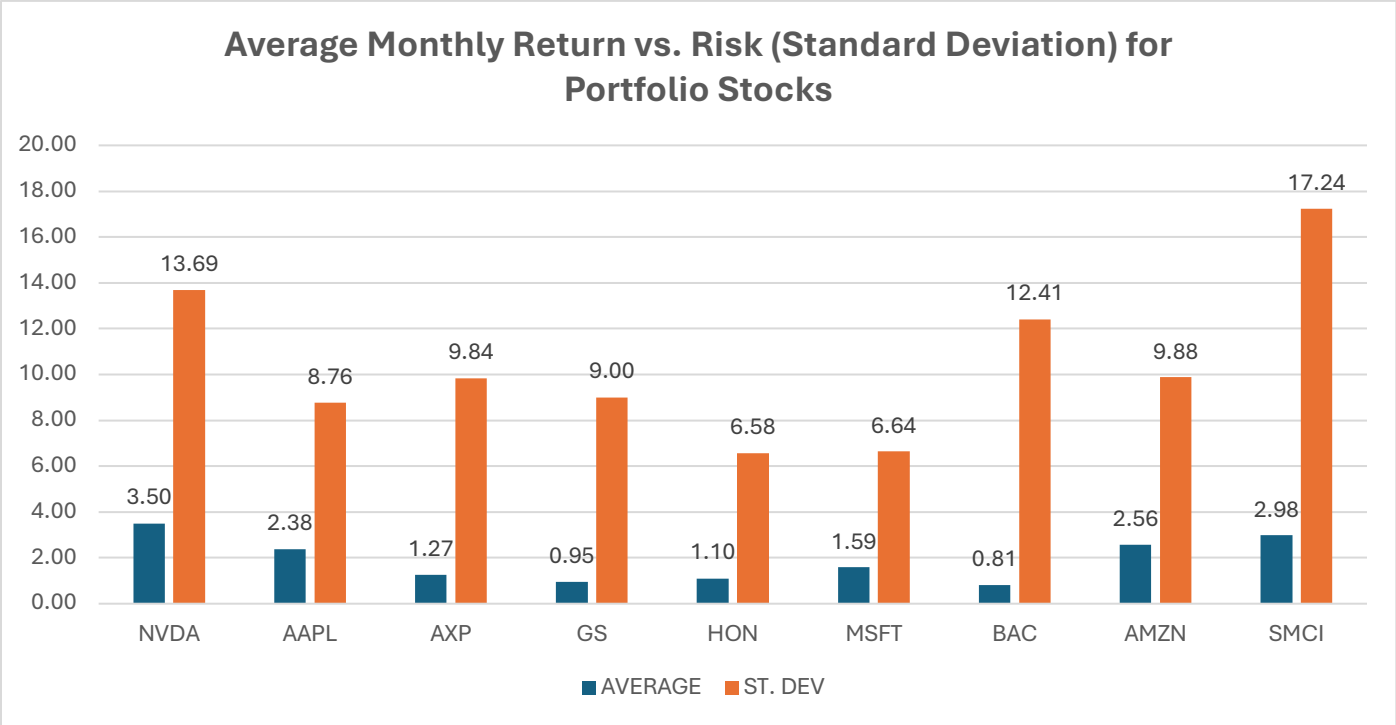
Quantitative Analysis

1. Average Return and Standard Deviation

To begin our quantitative analysis, we calculated the average monthly return (mean) and the standard deviation for each stock in the portfolio. The mean gives us a sense of the stock's overall performance by showing how much return it typically delivers in a month. On the other hand, the standard deviation helps us understand risk, as it measures how much the stock's return tends to vary — or in simple terms, how "bumpy" or unpredictable its performance is.

Stocks like NVIDIA (NVDA) and Super Micro Computer (SMCI) showed higher average returns, which makes them attractive for long-term growth. However, they also had high standard deviations, indicating greater volatility. In contrast, companies like Apple (AAPL) and Microsoft (MSFT) offered more moderate returns but came with relatively lower volatility, making them more stable choices.

The chart below visualizes this risk-return trade-off by comparing the mean and standard deviation for each stock. This helped us clearly identify which stocks offered the best return potential for the amount of risk involved — a key input into our portfolio construction process.



2. Monthly Return Distributions (Histograms)

While average return and standard deviation provide helpful summaries, they don't fully explain how returns are actually distributed over time. To explore this further, we used histograms to visualize the monthly return distribution of each stock. These charts show how often each stock experienced certain levels of gain or loss, allowing us to identify the shape, spread, and any unusual patterns in their return behavior.

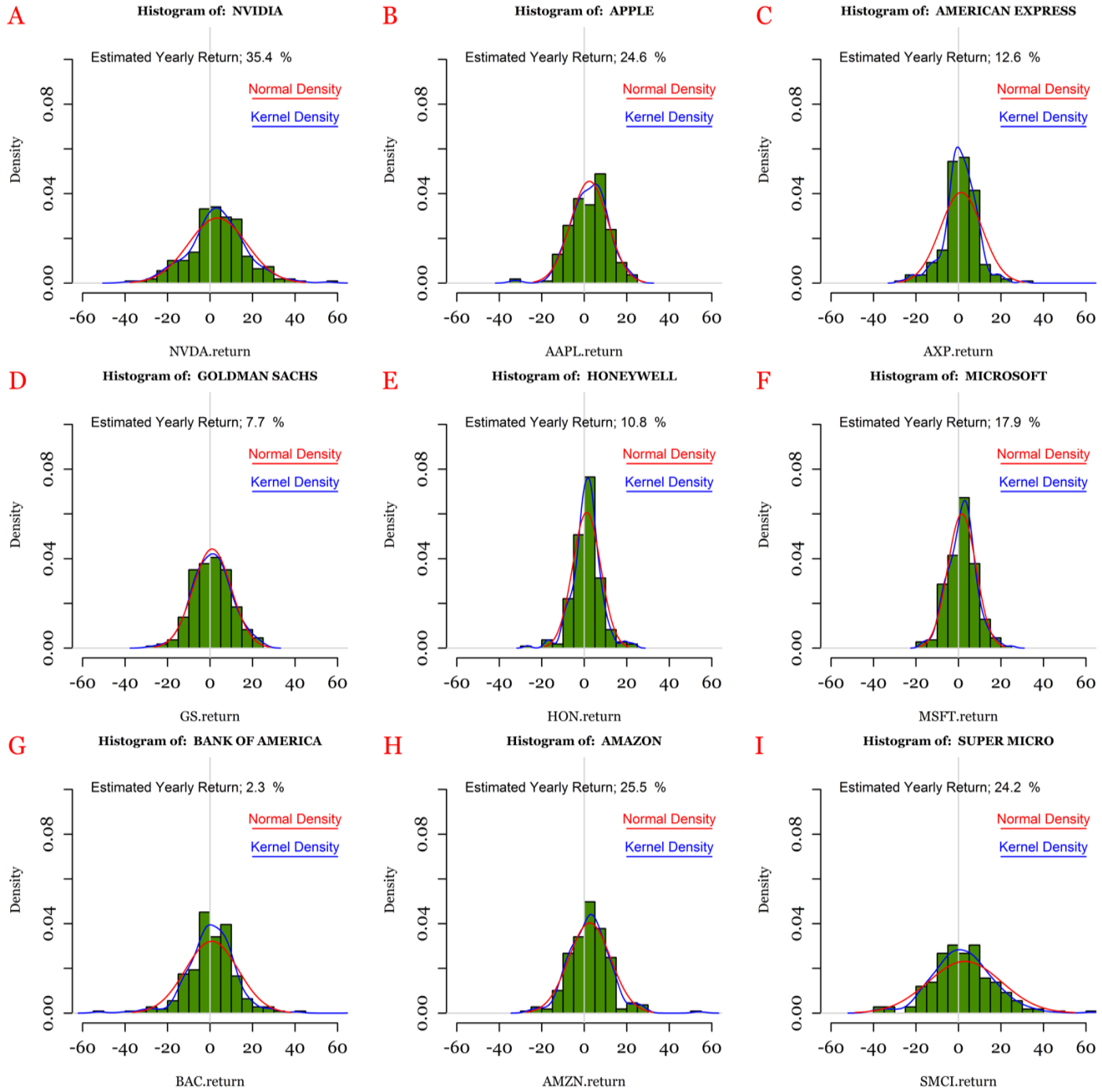
For example, stocks like NVIDIA and Super Micro have wide, flatter histograms, showing that their monthly returns vary greatly — from significant gains to sharp losses. This reflects high volatility and the possibility of both strong upside and downside. On the other hand, companies like Microsoft and Apple have more

concentrated histograms, with returns clustered around the average, indicating more predictable and stable performance.

Each histogram includes a blue kernel density curve, which provides a smooth estimate of the return distribution, and a red normal curve for comparison. In many cases, the real return patterns deviate from a perfect bell curve, especially for higher-risk stocks, showing that stock returns are not always normally distributed.

This analysis helped reinforce our understanding of individual stock behavior and supported our allocation decisions — especially in recognizing which assets carried higher downside risk and which offered more consistent performance.

Histograms of Monthly Rate of Returns (in %) of Nine Stocks

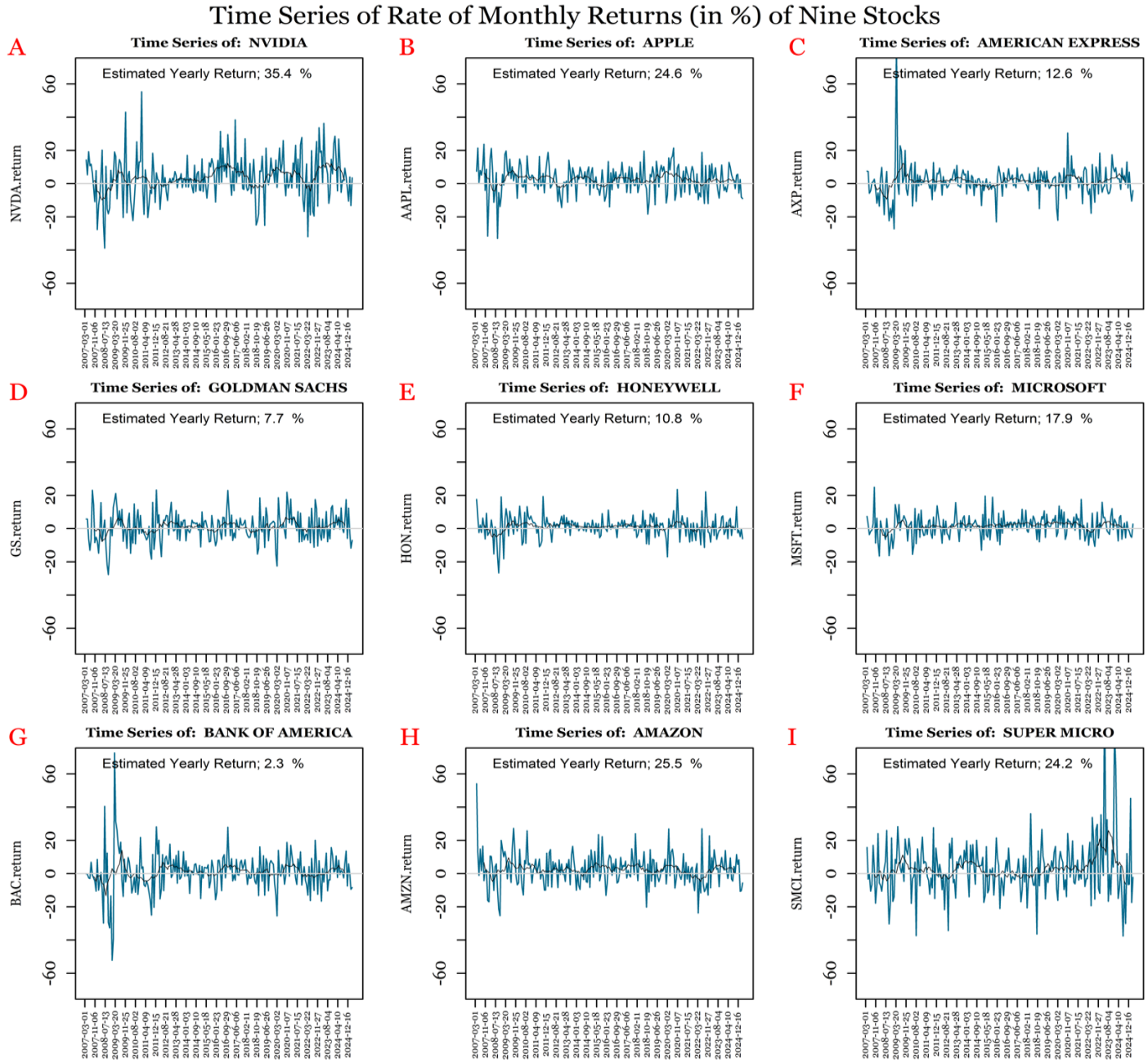


3. Time Series of Monthly Returns

To support the findings from our histogram analysis, we also examined time series graphs of monthly returns for each stock. These graphs offered a clearer view of how returns evolved over time, helping us identify periods of volatility, growth, and occasional outlier events.

For example, NVIDIA (NVDA), Super Micro (SMCI), and Amazon (AMZN) exhibited sharp upward and downward movements, reinforcing their status as high-risk, high-reward assets. Meanwhile, Apple (AAPL) and Microsoft (MSFT) showed more stable trends, suggesting lower volatility and consistent performance.

This analysis guided our portfolio design. In the 10-year strategy, we limited exposure to volatile stocks and emphasized risk-free assets. In the 30-year strategy, we leaned into growth stocks, allowing time to smooth out short-term fluctuations. Time series graphs helped us not only understand average return but also the consistency and risk timing behind it — crucial for building well-balanced, time-sensitive portfolios.



4. Portfolio Introduction: 10-Year and 30-Year Strategies

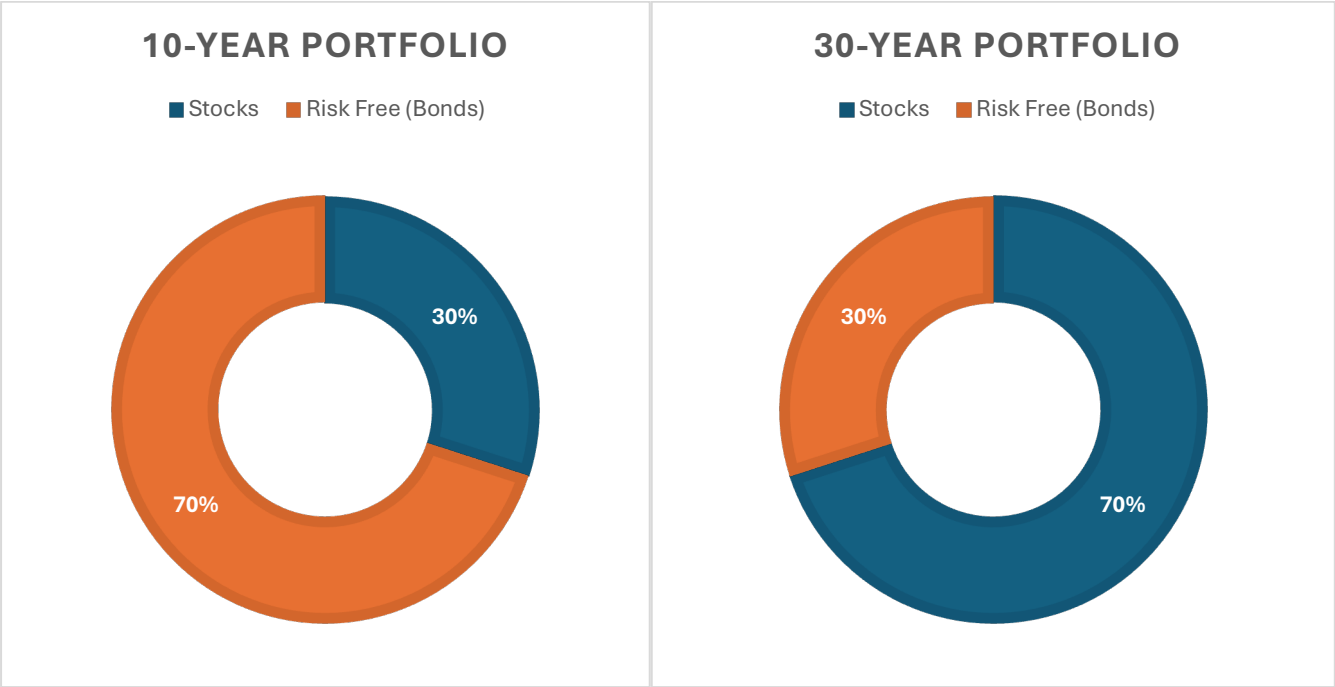
After analyzing each stock's historical return patterns and volatility, we constructed two portfolios based on different investment horizons: a **10-year strategy** focused on risk control, and a **30-year strategy** optimized for long-term growth. The goal was to use the return and risk insights from our previous analysis to build portfolios that align with the specific needs and timeframes of two types of investors — one with a shorter, more conservative horizon and another with a longer, growth-oriented view.

In the **10-year strategy**, we allocated **30%** of the investment into a carefully selected group of high-potential stocks — NVIDIA, Apple, Amazon, Microsoft, and Super Micro — while placing the remaining **70%** in risk-free assets such as bonds. This conservative mix was designed to protect against short-term volatility, especially since short selling was not allowed in this project. Without the ability to hedge through short positions, the bond component played a critical role in stabilizing returns.

For the **30-year strategy**, we took a more aggressive approach. Given the longer time horizon, the portfolio could afford to take on greater short-term volatility in exchange for higher long-term returns. Here, we allocated **70% to high-growth stocks** — NVIDIA, Apple, Amazon, and Super Micro — and only **30% to bonds**. This allowed us to maximize compounding gains while still keeping some downside protection.

These two portfolio structures were built using historical data, mean-variance analysis, and risk-return optimization, ensuring that each was tailored to its intended investment timeline and risk tolerance.

Allocation breakdown for the 10-year and 30-year investment strategies, balancing stocks and bonds based on risk tolerance and investment horizon.



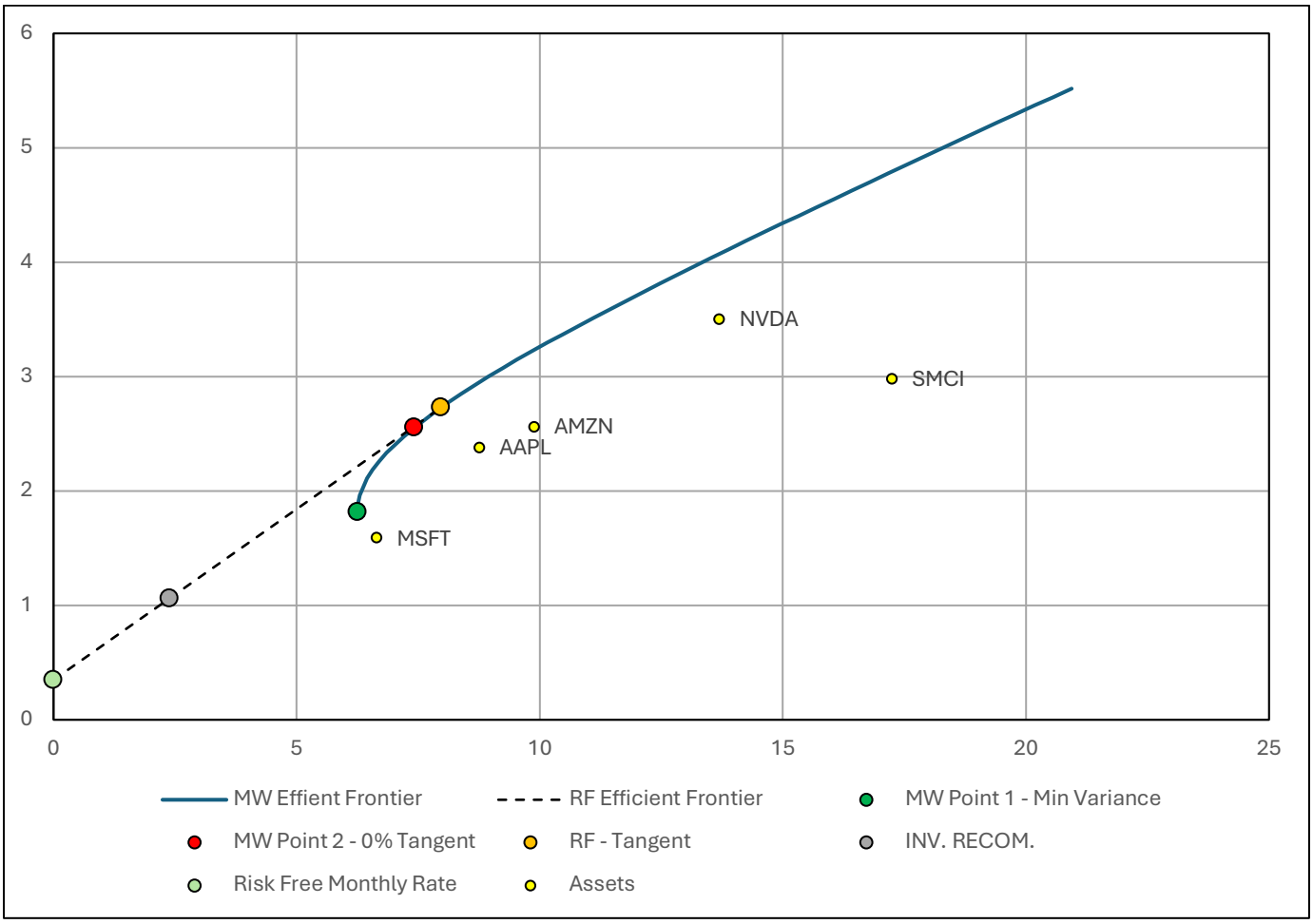
To evaluate how well these portfolio combinations balance risk and return, we applied mean-variance optimization techniques and plotted the Efficient Frontier and Capital Market Line, helping us identify the most efficient asset allocations under the constraints of each strategy.

5. Efficient Frontier and Capital Market Line

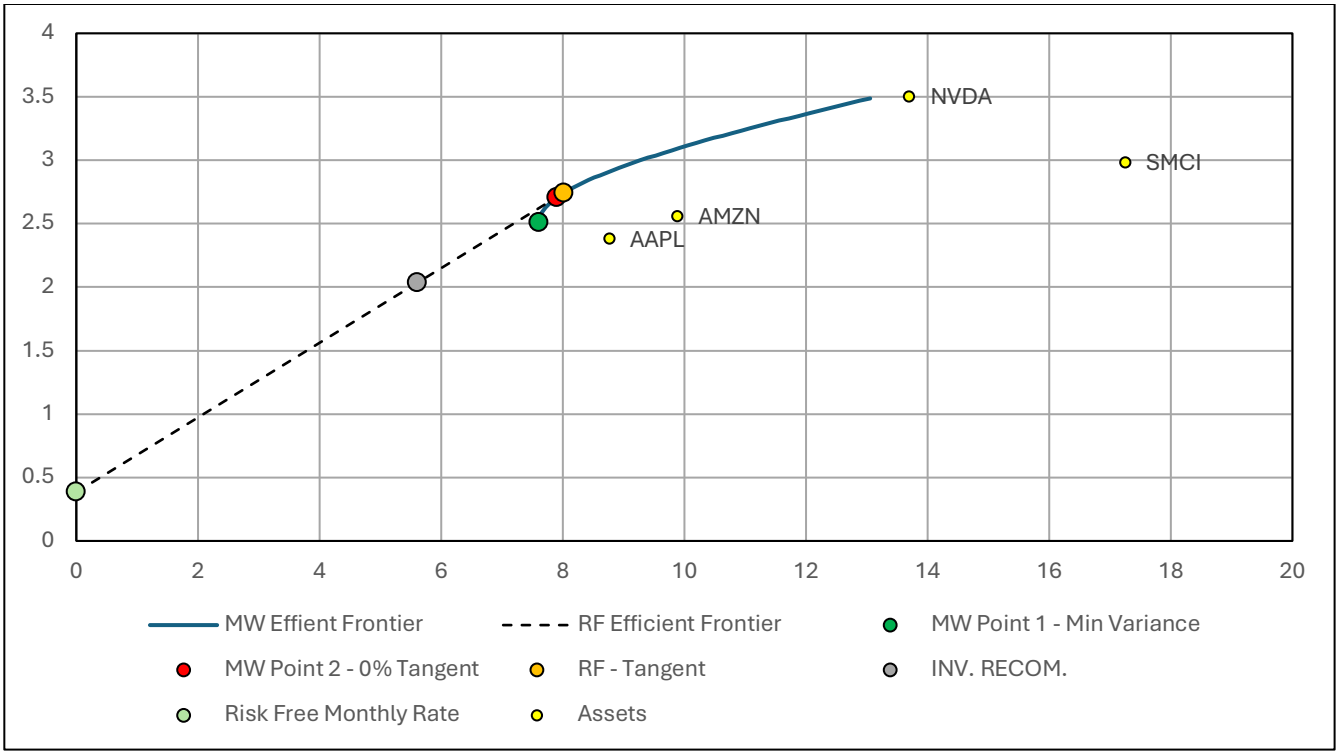
To assess the performance potential of the 10-year and 30-year portfolios, we applied mean-variance optimization using the Markowitz model to generate the Efficient Frontier — a curve that represents the best possible risk-return combinations available from the selected assets. Every point along this frontier corresponds to a portfolio that delivers the highest expected return for a given level of risk, or the lowest risk for a given level of return. Portfolios that fall below this curve are considered inefficient, as they take on more risk without offering better returns.

We created separate efficient frontiers for the 10-year and 30-year strategies using historical return data and the calculated standard deviations of each stock. The difference in shape and optimal allocation between the two curves reflects the time horizon and the acceptable level of volatility for each plan. In the 10-year model, the optimal portfolios leaned more toward the conservative end, while the 30-year frontier stretched further along the return axis, emphasizing higher growth opportunities despite greater risk.

To complement this, we plotted the Capital Market Line (CML) — a straight line that starts from the risk-free rate and is tangent to the efficient frontier. The point of tangency is known as the market portfolio, and it represents the ideal combination of risky assets when paired with a risk-free asset such as bonds. The slope of the CML indicates the Sharpe Ratio, or the return earned per unit of risk. A steeper slope reflects a more efficient portfolio.



(Efficient Frontier Graph and Capital Market Line for 10 Year Portfolio)



(Efficient Frontier Graph and Capital Market Line for 30 Year Portfolio)

Due to the no-short-selling constraint, all portfolios stayed within the feasible long-only region. This impacted the shape of the efficient frontier and restricted the use of leverage to improve returns. However, through careful asset selection and allocation, we were still able to identify portfolio combinations that lie on or near the efficient frontier — offering the best possible trade-off between risk and return under the project's constraints.

6. Value at Risk (VaR) and Conditional Value at Risk (CVaR)

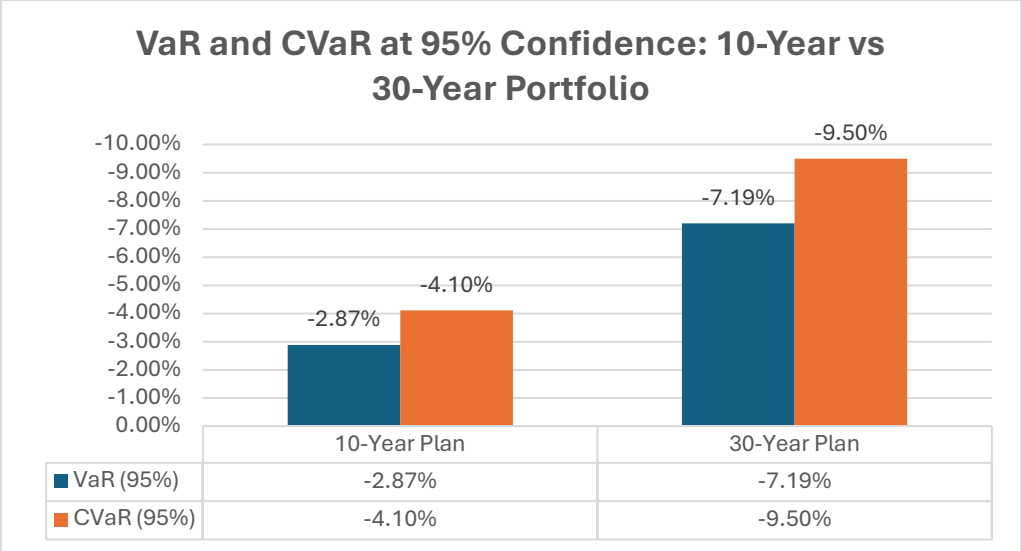
To further understand and quantify the potential downside of our portfolios, we calculated two important risk metrics: Value at Risk (VaR) and Conditional Value at Risk (CVaR) at a **95%** confidence level. These measures help estimate how much we might lose during particularly bad months, providing an additional layer of insight beyond standard deviation or historical trends.

VaR tells us the maximum expected loss in a given time period with **95%** confidence. For instance, in the 10-year portfolio, the VaR is approximately **-2.87%**, meaning there's only a **5%** chance that losses will exceed this amount in any given month. For the more aggressive 30-year portfolio, the VaR is around **-7.19%**, reflecting its higher exposure to volatile assets.

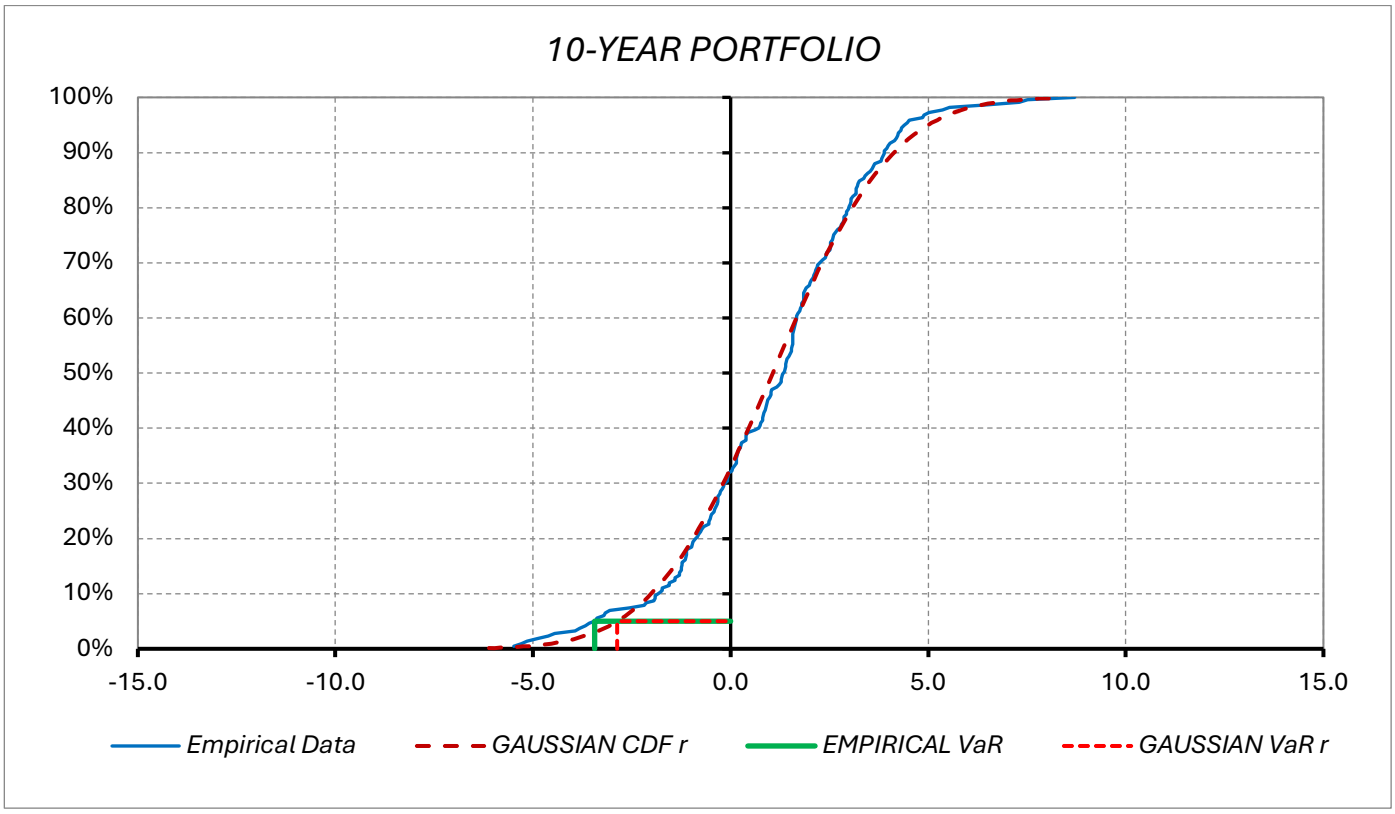
While VaR sets a loss threshold, CVaR (also known as Expected Shortfall) tells us how bad losses can get if that threshold is breached. In the 10-year strategy, the CVaR is about **-4.10%**, and in the 30-year plan, it's closer to **-9.50%**. These values represent the average loss in the worst **5%** of cases, offering a clearer view of potential extreme downside risk.

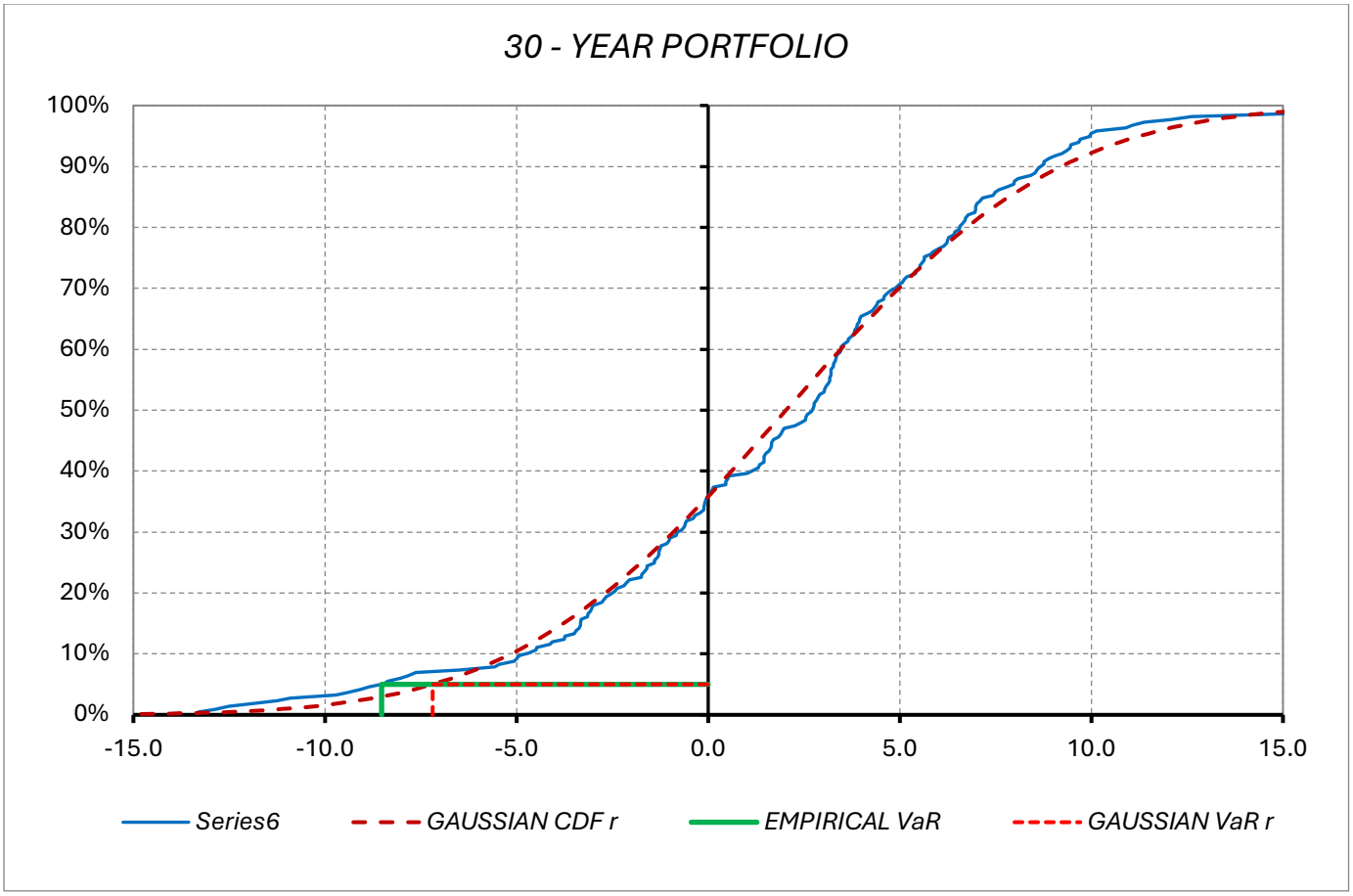
By comparing both VaR and CVaR across our portfolios and individual stocks, we confirmed that portfolios with higher allocations to growth stocks (like NVIDIA and Super Micro) are more exposed to significant

downside events. This analysis reinforced the importance of including risk-free bonds to cushion potential losses, particularly in the 10-year strategy where capital preservation is a priority.



The return distribution charts below for the 10-year and 30-year portfolios illustrate how monthly returns are spread over time, offering insights into the likelihood of various outcomes. In both charts, the histogram bars represent the actual distribution of historical returns, while the red line overlays the theoretical normal distribution. The blue kernel density curve provides a more accurate, data-driven view of the return patterns. For the 10-year portfolio, the distribution is tighter and more symmetric, indicating relatively lower volatility and fewer extreme outcomes. In contrast, the 30-year portfolio shows a wider spread with a heavier left tail, reflecting increased exposure to downside risk in pursuit of higher returns. These differences are critical to understanding the respective VaR and CVaR outcomes, as the 30-year strategy, while potentially more rewarding, also comes with greater risk of large losses in worst-case scenarios.





7. Sharpe Ratio and Jensen's Alpha

To evaluate how efficiently our portfolios are expected to perform relative to the risk they carry, we calculated two key performance metrics: the Sharpe Ratio and Jensen's Alpha.

The Sharpe Ratio measures how much excess return a portfolio generates for each unit of risk (standard deviation). It helps determine whether the return justifies the volatility. A higher Sharpe Ratio indicates a more attractive risk-adjusted performance. In our analysis, the 10-year portfolio had a lower Sharpe Ratio due to its conservative bond-heavy allocation, while the 30-year portfolio showed a higher Sharpe Ratio, reflecting better long-term return potential despite greater volatility. These results are consistent with their positions relative to the Capital Market Line, where steeper slopes indicate more efficient portfolios.

We also calculated Jensen's Alpha, which measures a portfolio's excess return above what CAPM would predict, given its risk (beta) and the market return. A positive alpha indicates that the portfolio is outperforming the market on a risk-adjusted basis. Our findings showed that both portfolios produced a small but positive alpha, suggesting that the selected asset combinations have potential to beat the market return expectations — especially the 30-year strategy, which benefits from long-term growth-focused stocks.

Together, these metrics provided strong evidence that both portfolios are not only well-balanced in terms of risk and return, but are also positioned to deliver above-market performance within their respective risk tolerances.

This project aimed to design two data-driven investment strategies tailored to different time horizons: a risk-conscious 10-year plan and a growth-oriented 30-year plan. Through detailed quantitative analysis, we evaluated each stock's historical return and volatility using mean and standard deviation, and visualized their behavior using histograms and time series plots. These insights formed the foundation for selecting assets that offered strong return potential while balancing risk appropriately.

Using mean-variance optimization, we constructed the Efficient Frontier for both portfolios, identifying optimal asset combinations based on their risk-return profiles. The Capital Market Line further confirmed the efficiency of our selections, showing how the inclusion of a risk-free asset (bonds) improved portfolio performance, especially in the absence of short-selling.

To assess performance under extreme conditions, we calculated Value at Risk (VaR) and Conditional Value at Risk (CVaR) at a 95% confidence level, confirming that the portfolios remain within acceptable loss thresholds — particularly important for the 10-year plan. We also evaluated overall portfolio efficiency through the Sharpe Ratio and Jensen's Alpha, both of which indicated solid risk-adjusted performance and a positive ability to outperform market expectations.

While the quantitative analysis provided a detailed breakdown of risk, return, and optimization strategies, it's equally important to view these portfolios through a practical lens. To reinforce how the 10-year and 30-year strategies align with real-world needs, the following investor profiles illustrate how each portfolio can be tailored to different financial goals, time horizons, and risk preferences. This bridges the analytical findings with the individuals they're designed to serve.

Profile Attribute	10-Year Portfolio Investor	30-Year Portfolio Investor
Goal	Preserve capital and generate steady monthly returns	Maximize long-term wealth through compounding
Time Horizon	Medium-term (e.g., retirement or home purchase in 8–12 years)	Long-term (e.g., early-career investor with 25–30 years of investing runway)
Risk Tolerance	Low to Moderate — prefers safety and predictability	Moderate to High — accepts short-term volatility for higher long-term gains
Investment Strategy	30% in high-quality growth stocks, 70% in bonds for stability	70% in high-growth stocks, 30% in bonds for baseline security
Expected Return	0.80% – 1.33% monthly	1.41% – 2.66% monthly
Downside Risk	VaR: -2.87%, CVaR: -4.10%	VaR: -7.19%, CVaR: -9.50%

In conclusion, this report demonstrates how combining historical data analysis with financial theory allows for the construction of tailored investment strategies that are not only statistically optimized but also practical under real-world constraints. Both portfolios reflect a careful balance of return, risk, and time horizon — offering strong frameworks for medium- and long-term wealth generation.

Investment Strategies

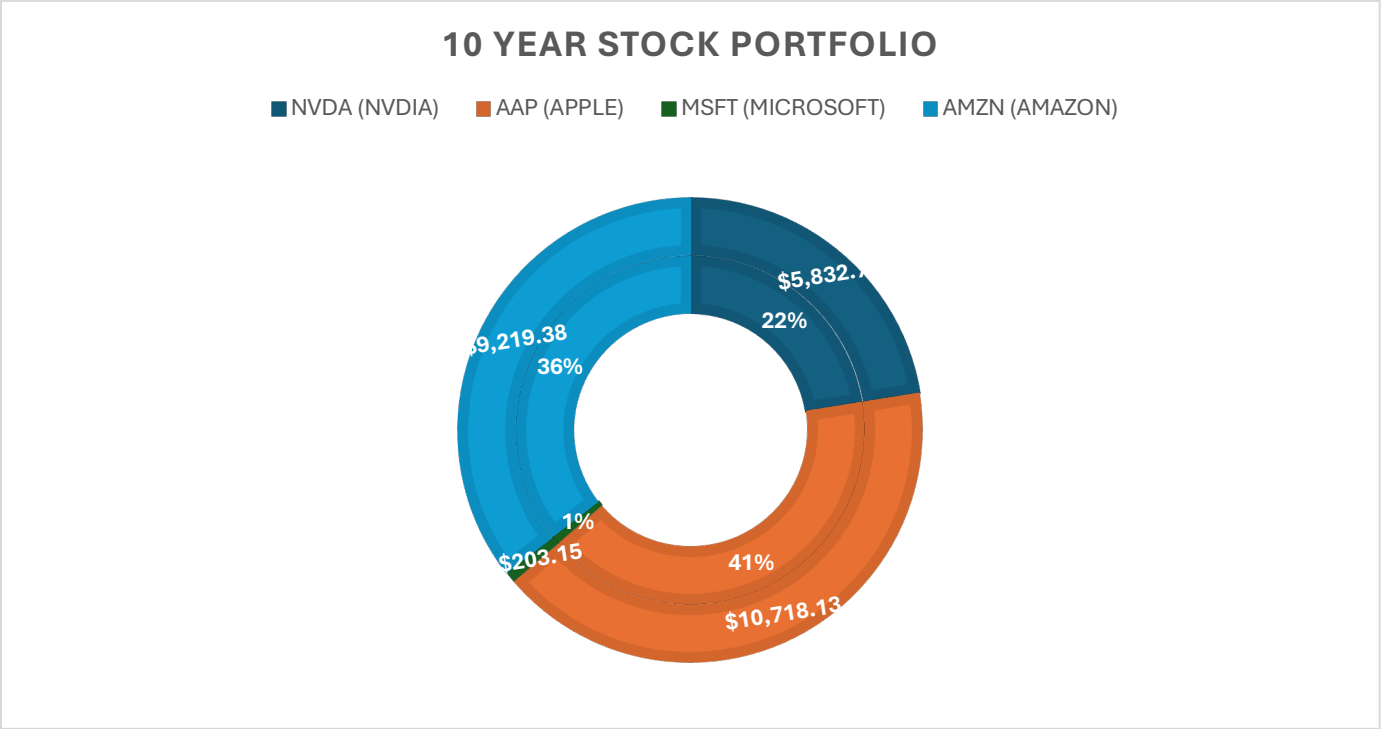
We have structured our investment strategy around two-time horizons: 10 years and 30 years. By dividing the portfolio this way, we can align investment choices with different financial goals, balancing short- to mid-term growth with long-term wealth accumulation.

1. 10 Year Portfolio

For the 10-year duration of the portfolio, short selling is not allowed; we are restricted to only buying stocks. To address this, we analyzed various combinations of stocks and finalized a set that offers the best possible risk-to-reward ratio. By "risk-to-reward ratio," we mean that achieving a portfolio with 0% risk is nearly impossible without shorting. Therefore, we selected a combination where the risk is minimized, and the potential rewards are maximized relative to the risk taken.

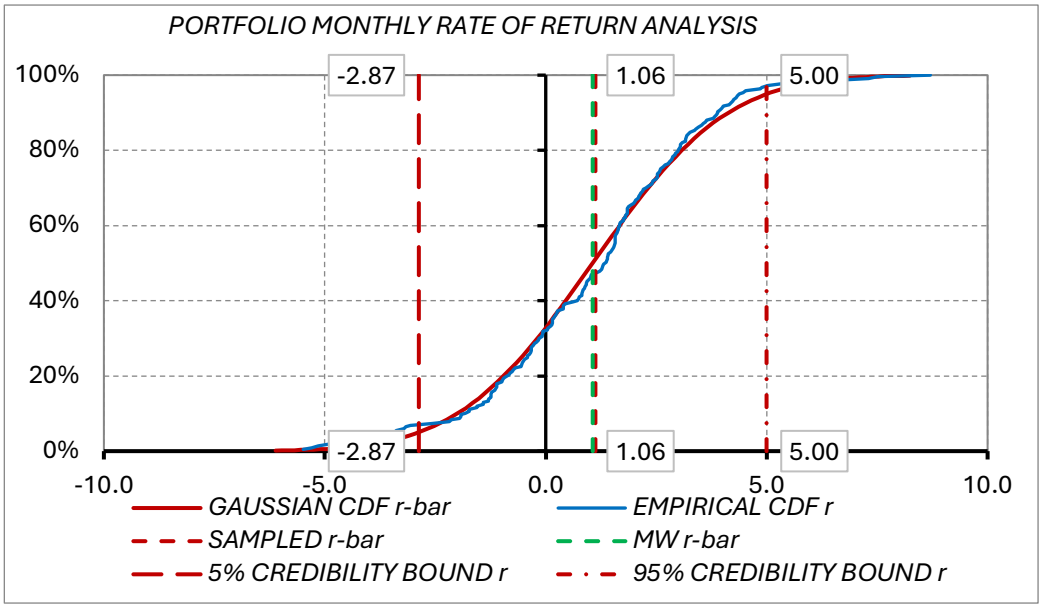
Our analysis concluded that allocating 30% of the funds to high-potential growth stocks — specifically NVIDIA (NVDA), Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), and Super Micro Computer (SMCI) where the portfolio weights have been mentioned below — and the remaining 70% to risk-free investments like bonds would optimize returns.

STOCKS	**ABS(WEIGHT)**	**AMOUNT**
NVDA (NVDIA)	19.442 %	$ 5,832.72
AAP (APPLE)	35.727 %	$ 10,718.13
MSFT (MICROSOFT)	0.677 %	$ 203.15
AMZN (AMAZON)	30.731 %	$ 9,219.38
SMCI (SUPER MICRO COMPUTER)	13.422 %	$ 4,026.62
TOTAL	100 %	$ 30,000.00

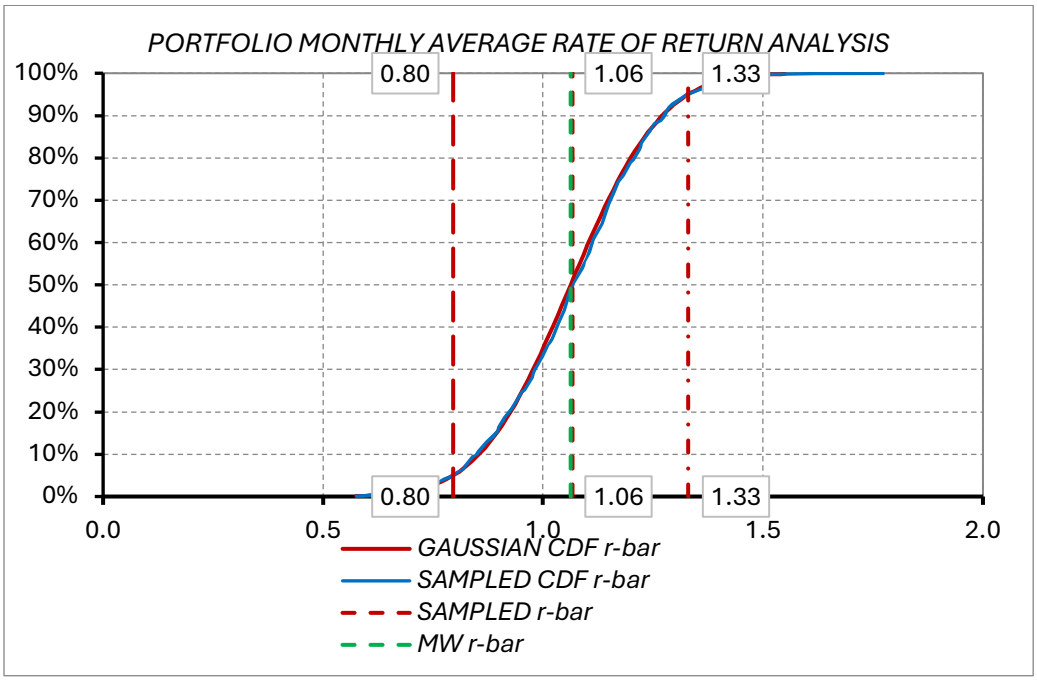


Based on our deep research, we are 95% confident that:

1. Each month, the portfolio could see returns ranging between -2.87% (a loss) and +5% (a gain), achieving a risk-to-reward ratio of approximately 1:2.

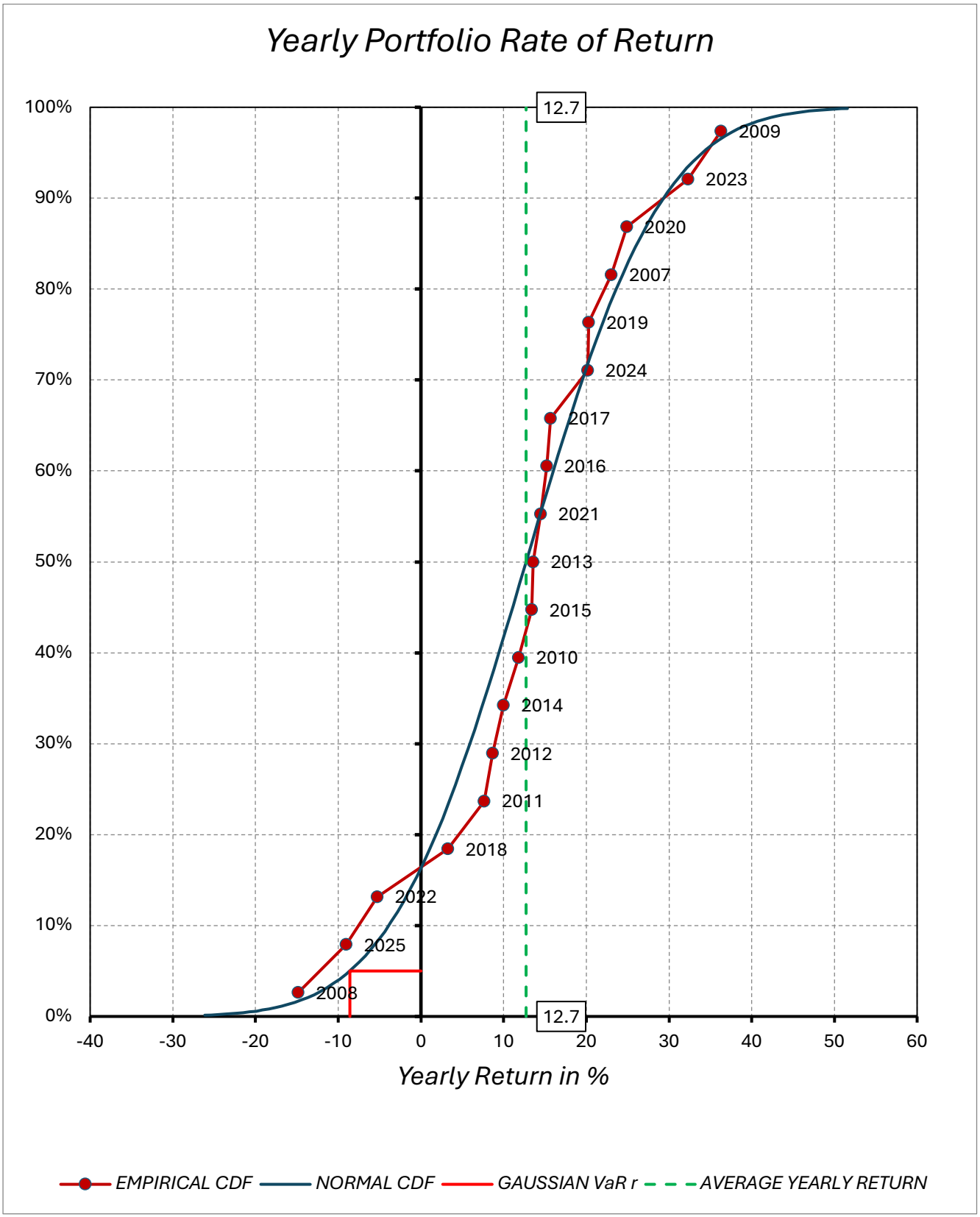


2. Over 10 years, considering the low likelihood of consistent losses over consecutive periods, the average monthly return is projected to range between 0.80% and 1.33% in profit.



Thus, even in the most conservative scenario, the quantitative analysis shows that the expected average return remains positive for the selected stock combination.

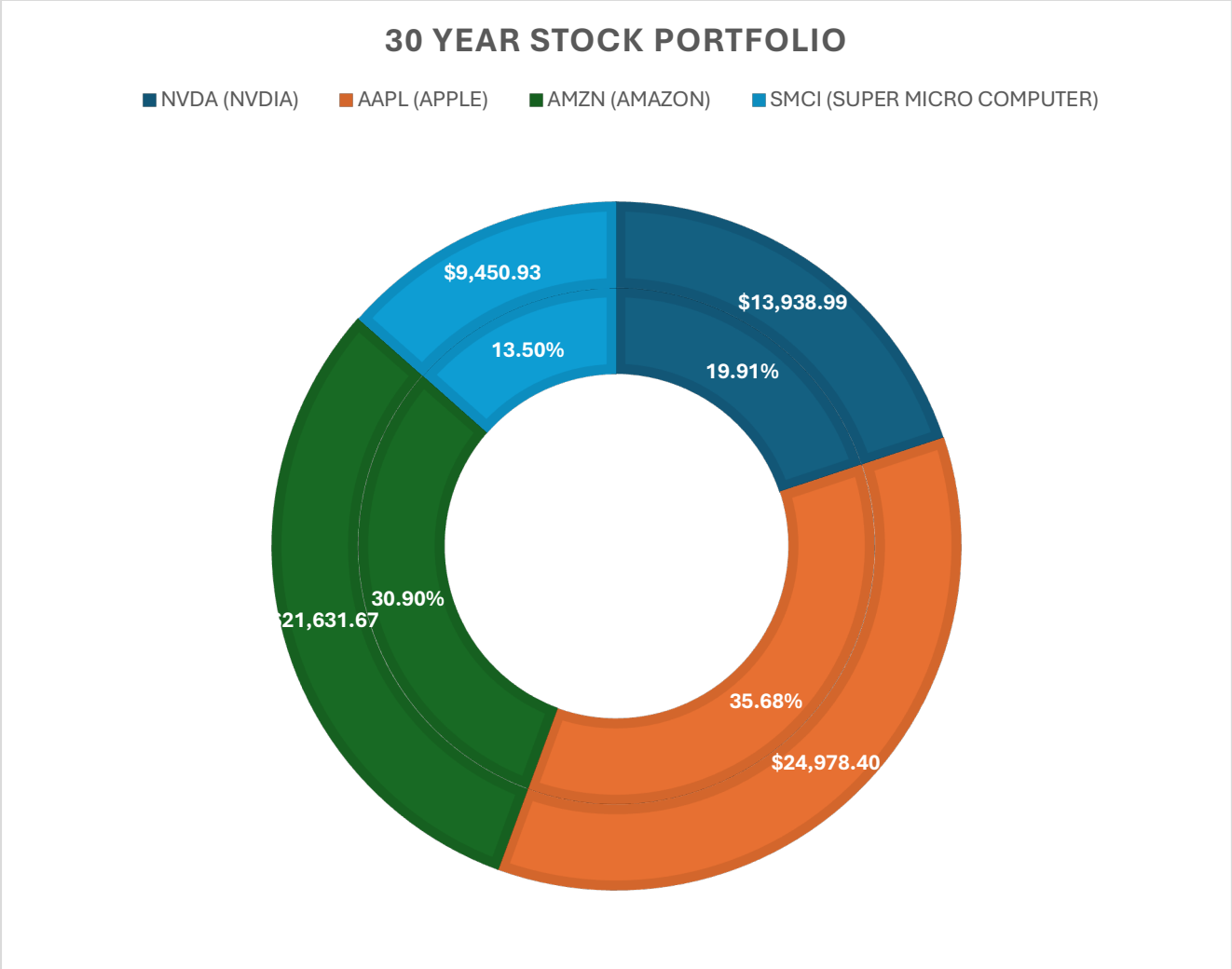
To illustrate the expected return consistency of the 10-year portfolio, the following chart shows its annual rate of return across the evaluation period. The results reflect a more conservative growth path with reduced volatility, aligning with the portfolio's lower-risk investment approach.



2. 30 Year Portfolio

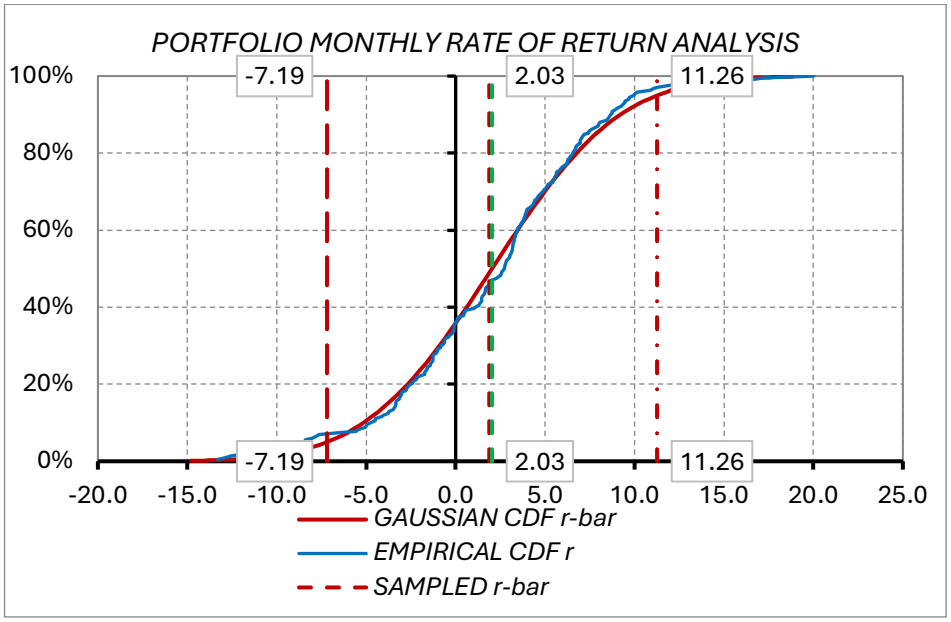
For the 30-year portfolio, we followed the same conditions as the 10-year plan, where only stock buying is allowed and short selling is not permitted. After extensive quantitative research, we selected a combination of four stocks: NVIDIA (NVDA), Apple (AAPL), Amazon (AMZN), and Super Micro Computer (SMCI). In this strategy, 70% of the investment is allocated to these stocks, while the remaining 30% is placed in risk-free assets like bonds to reduce overall risk.

STOCKS	**ABS(WEIGHT)**	**AMOUNT**
NVDA (NVDIA)	19.913 %	$ 13,938.99
AAPL (APPLE)	35.683 %	$ 24,978.40
AMZN (AMAZON)	30.902 %	$ 21,631.67
SMCI (SUPER MICRO COMPUTER)	13.501 %	$ 9,450.93
	100 %	$ 70,000.00

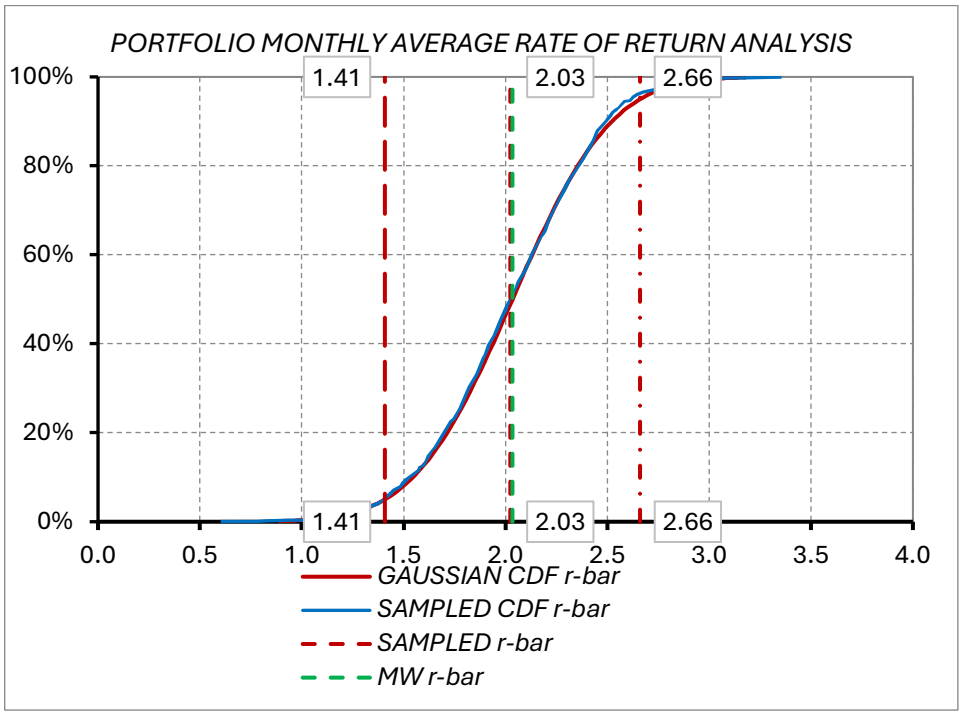


Based on our analysis, we are about 95% confident in the following outcomes:

1. Each month, the portfolio's returns are expected to vary between a 7.19% loss and an 11.26% gain.

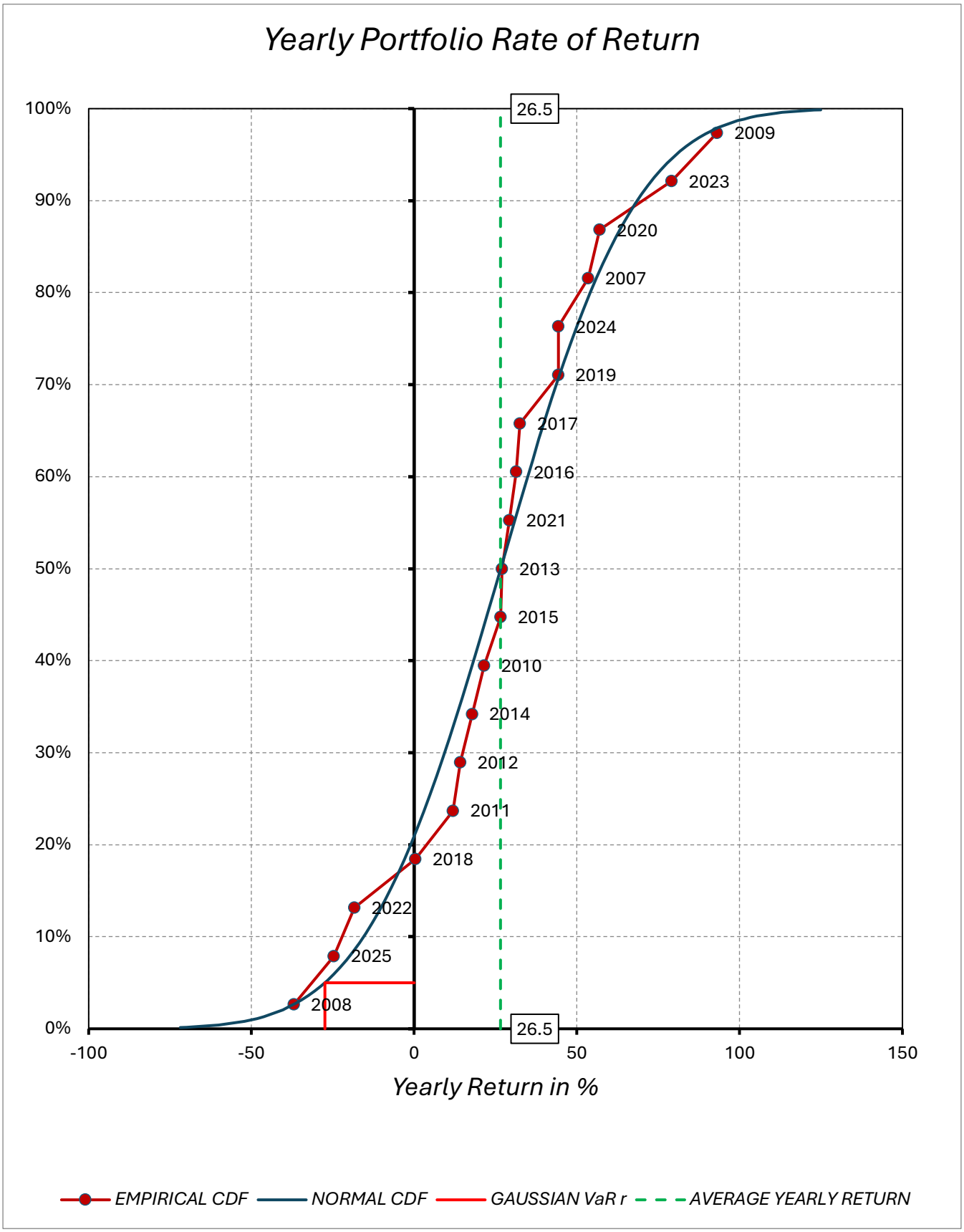


2. Over the 30-year period, the average monthly return is projected to range between 1.41% and 2.66%, showing strong potential for long-term growth.



This setup balances higher exposure to growth stocks with the stability provided by bonds, aiming to maximize returns while managing risk over a longer time horizon.

The chart below presents the annual rate of return for the 30-year portfolio. This strategy, designed for long-term growth, displays higher year-to-year variability, consistent with its aggressive allocation and higher return potential over time.



Assumptions And Limitations

This portfolio analysis is based on several key assumptions to simplify the modeling process and focus on evaluating risk and return. First, it assumes that the historical monthly returns of the selected stocks reflect future trends, allowing us to estimate expected returns and risk using average returns (mean) and standard deviation. We also assume that these returns are normally distributed and occur independently each month, without being influenced by prior outcomes. All investment positions are long-only, since short selling and leverage were not permitted in the scope of this project. Bonds are treated as stable, risk-free assets with consistent performance. It is further assumed that the asset allocation (e.g., 30% stocks and 70% bonds for the 10-year plan) remains fixed over time without rebalancing. Real-world factors like taxes, transaction costs, and inflation are excluded for simplicity. Based on the statistical findings and return range modeling, we can be approximately **95% confident** in the projected performance ranges of the portfolios under normal market conditions.

Despite its usefulness, the analysis has several limitations that must be considered. Since it relies entirely on historical data, it may not fully capture future market changes, unexpected global events, or economic crises. Standard deviation is used as the sole measure of volatility, which may not reflect risks from rare or extreme scenarios. Broader economic indicators like interest rates, inflation, or political shifts were not incorporated into the model, though they can strongly affect investment performance. The analysis also focuses only on a small group of U.S.-based growth stocks and excludes other potentially beneficial asset classes such as international equities, real estate, or commodities. While we are 90% confident in the model's outputs under typical conditions, the results may differ due to these factors.

Risk Management Approach

Since short selling was not allowed in this project, we couldn't use it to protect the portfolio from falling stock prices. Instead, we managed risk by carefully choosing stocks and deciding how much money to put in each part of the portfolio. We used Excel to check how risky each stock was by looking at how much its returns went up and down in the past. For the 10-year strategy, we put only 30% of the money in riskier growth stocks like NVIDIA and Super Micro, and the other 70% in safer investments like bonds. This helped keep the portfolio more stable in the short term. In the 30-year strategy, we increased stock exposure to 70%, because a longer time frame gives more room for recovery and growth. Overall, our goal was to balance risk and return by using smart allocation, since short selling wasn't an option.

Conclusion

In a world of uncertainty, building wealth requires more than chasing returns — it demands a balance between ambition and protection. Through data-driven analysis, disciplined asset selection, and carefully structured allocation, this project delivers two portfolios that reflect not only strong return potential but also a deep respect for risk. Whether it's the capital-preserving focus of the 10-year strategy or the compounding power of the 30-year approach, each portfolio is backed by thorough research, tested through market data, and optimized to serve investors with distinct goals.

This isn't just a plan — it's a roadmap built on logic, not luck. The numbers support it. The strategy defends it. And the vision behind it invites trust.

"Investing should be more like watching paint dry or watching grass grow. If you want excitement, take $800 and go to Las Vegas."
— Paul Samuelson (Nobel Winning Economist)